

CS193 — Week 3 Homework

Exploring Real Weather Data from the Command Line

INSTRUCTOR ANSWER KEY

Data source: The WHIN (Wabash Heartland Innovation Network) weather network operates a dense set of weather stations across a 10-county region of north-central Indiana. Your files hold a snapshot of readings from 24 of those stations.

Time: 15–20 minutes • **Reading:** *The Linux Command Line* (linuxcommand.org/tlcl.php) — the chapters on navigation and I/O redirection.

Skills: finding files (`find`, `ls`) and pipes & filters (`head`, `cut`, `grep`, `sort`, `uniq`, `wc`).

Setup

Download the **cs193_week3** folder. Inside you'll find:

- `whin_stations.csv` — one row per station (the current snapshot)
- `station_logs/` — one file per station, with hourly readings

Open a terminal and move into the folder before you start:

```
cd cs193_week3
```

Part A — Finding your way around

Every station writes its own log file. Let's locate them.

A1. List the log files, then count them. How many stations logged data today?

```
ls station_logs
ls station_logs | wc -l
```

```
24  (one log file per station)
```

A2. Now count every CSV file in the folder, including the snapshot. `find` searches inside sub-folders for you.

```
find . -name "*.csv" | wc -l
```

```
25  (24 station logs + whin_stations.csv)
```

A3. WHIN's Tippecanoe County stations all start with TIP. List just those log files.

```
find station_logs -name "TIP*"
```

```
station_logs/TIP01_2026-07-11.csv
station_logs/TIP02_2026-07-11.csv
station_logs/TIP03_2026-07-11.csv  (3 files)
```

Part B — Reading the data with pipes & filters

Now work with the snapshot file `whin_stations.csv`. The columns are: `station_id`, `station_name`, `county`, `latitude`, `longitude`, `temp_f`, `humidity_pct`, `wind_mph`, `rain_in`, `soil_moist_pct`, `observed_at`.

B1. Peek at the first 5 lines. What does the very first line contain?

```
head -5 whin_stations.csv
```

The first line is the **header row** (the column names). The next 4 lines are station readings.

B2. How many stations are in the snapshot? Remember: the header row is not a station, so skip it with `tail -n +2` ("start at line 2").

```
tail -n +2 whin_stations.csv | wc -l
```

24

B3. Print just the station names. They live in column 2, and the file is comma-separated, so tell `cut` the delimiter is a comma.

```
cut -d, -f2 whin_stations.csv
```

station_name
Fowler
Oxford
Delphi
... (all 24 names)

B4. How many stations are in **White** County? `grep -c` counts matching lines.

```
grep -c White whin_stations.csv
```

3

B5. (*Students write this command themselves.*) Count how many stations are in *each* county: take the county column, sort it, then collapse duplicates with a count.

```
tail -n +2 whin_stations.csv | cut -d, -f3 | sort | uniq -c
```

```
2 Benton    2 Carroll   3 Cass      3 Clinton   2 Fountain
2 Montgomery 2 Pulaski   3 Tippecanoe 2 Warren    3 White
```

B6. (*Students write this command themselves.*) Which station is the **coolest** in this snapshot? Temperature is column 6. Sort the rows numerically by that column and take the first one.

```
tail -n +2 whin_stations.csv | sort -t, -k6 -n | head -1
```

FOU02,Covington,Fountain,...,80.0,... → **Covington** (Fountain County), 80.0°F

Bonus (optional, not graded)

Which station is the **hottest on average**? Each station's hourly readings are in its own file in `station_logs/`. Look across all the logs, average each station's temperatures, then sort to find the highest.

```
awk -F, 'FNR>1{s[FILENAME]+=$2;n[FILENAME]++} END{for(f in s) print s[f]/n[f], f}' station_logs/*.csv | sort -n | tail -1
```

76.58 station_logs/WHI03_2026-07-11.csv → **WHI03 (Reynolds, White County)**, ~76.6°F average

Ungraded stretch. Accept any correct approach --- a for loop over the files works too. The point is aggregating across files and sorting.